Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/25/2007
PI Name:	Pollock, Neal Ph.D.		
Project Title:	Use of Exercise and Partial Decompression to Reduce Oxygen Prebreathe Requirements for 4.3 PSIA Extravehicular Activity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	neal.pollock@duke.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919-668-0032
Organization Name:	Duke University Medical Center		
PI Address 1:	Center for Hyperbaric Medicine and Environmental Physiology		
PI Address 2:	P.O. Box 3823		
PI Web Page:	http://hyperbaric.mc.duke.edu/		
City:	Durham	**State:**	NC
Zip Code:	27710	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/01/2006	**End Date:**	03/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 3/31/2008 per P. Denkins/JSC (6/07) NOTE: added to Task Book in 6/2007 when first learned of it from JSC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ06HD74A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts, like divers, are exposed to changes in pressure that can lead to decompression sickness (DCS), otherwise known as "the bends". When an astronaut leaves the Space Station for Extravehicular Activity (EVA), the pressure of the space suit (4.3 psia) is one third of sea level pressure (14.7 psia), or equivalent to the pressure at an altitude of 30,300 feet. Astronauts breathe oxygen prior to an EVA ('prebreathe') to eliminate nitrogen from the body, thus decreasing the risk of developing decompression sickness. This study is designed to determine the safest procedure for protecting astronauts from DCS during the many spacewalks required for construction and maintenance of the International Space Station. Extravehicular activity is very demanding, both physically and mentally, partially due the nature of the space suit. The space suit currently in use consists of a Hard Upper Torso with flexible extremities. The resting position or maximum volume of a spacesuit is found with the arms elevated overhead in a relaxed position. Astronauts work to alter the resting position of the spacesuit in addition to the work required for the EVA. This study endeavors to simulate both the effects of the space environment on the body and the effort of spacesuit/EVA activity. The specific goals of this project are to: Measure the oxygen consumption necessary to complete prebreathe protocols during astronaut simulation activity. Determine the operational suitability of a 1.5 hr in-suit prebreathe protocol. Provide scientific understanding of the relationship between intermittent exercise intensity and duration and the occurrence of DCS and VGE.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2006; no progress report received during this reporting period.
Bibliography Type:	Description: (Last Updated: 03/27/2017)